FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY

UCCD3084 Graphics Programming For Extended Reality

TITLE: Fire Extinguisher Training VR Simulation in a Factory

Group 6: LoveUtar

Submission Date: 19/4/2026

Name	Student ID	Course	Practical Group
Mak Zi Qian	2106110	CS	P2
Ong Ti Xuan	2104503	CS	P2
Yip Siew Hui	2104912	CS	P2

Table of Contents

Project Scope 3

Environment & Hazard Placement 3

Visual Diagram 4

System Design Explanation 5

XR Interaction Design 6

Screenshots 7

Team Contribution 12

Project Scope

This project aims to develop a VR training simulation that teaches users how to properly operate a CO_2 fire extinguisher in a factory environment. The system is designed based on the P.A.S.S. method, which includes Pull, Aim, Squeeze, and Sweep.

The simulation allows users to interact with a fire extinguisher using XR interaction techniques such as grabbing, aiming, and triggering. The system provides step-by-step guidance through UI prompts to ensure that the user follows the correct procedure.

Several key features are implemented in this project. The fire extinguisher is designed as an interactive object with grabbable functionality. The system includes safety pin removal for the Pull step, a direction-based aiming system for the Aim step, and spray activation using input control for the Squeeze step. In addition, sweep detection is implemented to ensure the user performs the correct motion to extinguish the fire effectively. The fire simulation responds visually to user actions, and a user interface provides real-time prompts and feedback throughout the training. A reset function is also included to allow users to restart the simulation easily. Two user modes: Training Mode (guided step-by-step P.A.S.S. instructions) and Free Roam Mode (unrestricted exploration and practice in the factory environment)

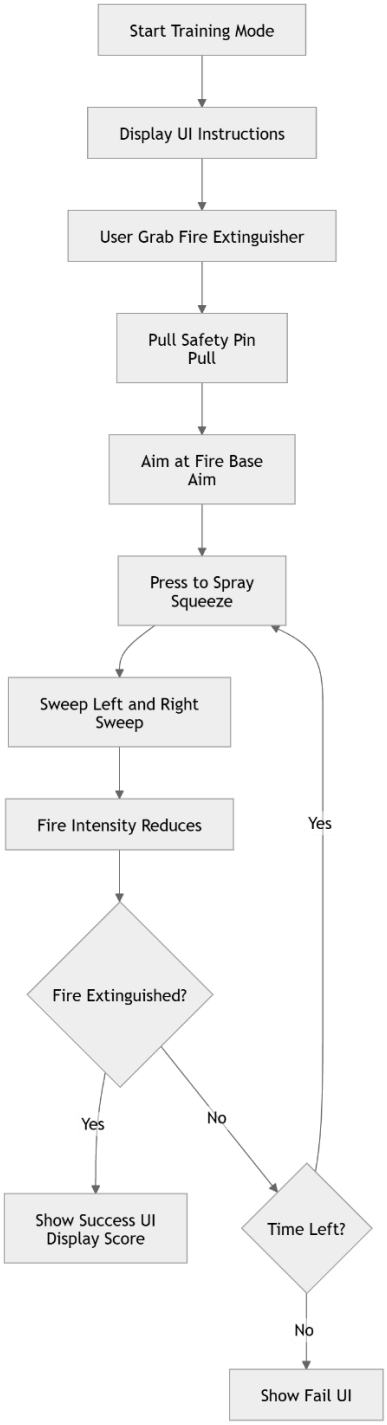
The project focuses on creating an intuitive and realistic training experience while ensuring usability and user comfort in a VR environment.

Environment & Hazard Placement

The simulation uses a factory scene as the base environment. The fire hazard is placed in a designated zone within the factory, representing an electrical fire scenario. Safety signs, CO_2 identification labels, and emergency exit signs are added to enhance realism and improve the overall training experience. The factory environment supports both guided Training Mode and open Free Roam Mode, allowing users to learn and practice fire safety procedures in a flexible VR experience.

Visual Diagram

The diagram illustrates the overall process of the VR fire extinguisher training system. The user begins by entering the training mode and receiving instructions through the user interface. The user then grabs the fire extinguisher and follows the P.A.S.S. method, which includes pulling the safety pin, aiming at the base of the fire, activating the spray, and performing a sweeping motion. The system continuously checks the fire intensity and timer. If the fire is successfully extinguished within the time limit, a success screen is displayed along with the score. Otherwise, the user may continue the process or receive a failure screen when time runs out.



System Design Explanation

The system is developed using Unity with XR Interaction Toolkit. The simulation consists of several main components including the fire extinguisher object, fire system, user interface, and user interaction controls.

The fire extinguisher is designed as an interactive object that users can grab and manipulate. The safety pin must be removed before the spray function is activated, ensuring correct procedural training based on real-world fire safety practices.

The fire system reacts dynamically to user actions. When the spray is correctly aimed at the base of the fire and applied continuously, the fire gradually decreases in intensity until it is fully extinguished. If the user does not perform a proper sweeping motion while spraying, the fire intensity will increase again, reinforcing correct usage technique.

The user interface provides real-time instructions and feedback, guiding the user through each step of the P.A.S.S. method (Pull, Aim, Squeeze, Sweep). This ensures that the learning experience is structured, clear, and easy to follow.

In addition, the system includes keyboard-based interaction controls to support simulation testing without VR hardware. Users can enable or disable the UI display, grab and release objects, switch movement modes, and perform spraying actions using specific keyboard inputs. This allows the simulation to function both in VR mode and desktop simulation mode for development and demonstration purposes.

XR Interaction Design

The XR interaction design in this project focuses on creating a natural and intuitive user experience in a virtual reality environment. The system uses the XR Interaction Toolkit in Unity to allow users to interact with objects through hand-based controls and controller inputs.

Users can directly grab the fire extinguisher using the VR controllers, simulating real-world hand interaction. The grabbing mechanism allows the object to follow the user's hand movement smoothly, making the interaction feel realistic. The system also supports two-hand interaction, where the user uses one hand to hold the extinguisher and the other hand to interact with specific components such as the safety pin.

The interaction design follows the P.A.S.S. method step by step. First, the user must pull the safety pin by moving their hand close to the pin area and triggering the interaction. Next, the user aims the extinguisher at the base of the fire using hand direction. The aiming system is based on the forward direction of the controller, allowing accurate targeting. The user then presses the input button to activate the spray function, simulating squeezing the handle. Finally, the user performs a sweeping motion from left to right to extinguish the fire effectively.

Additional keyboard inputs are included for testing and simulation purposes. For example, the user can press specific keys to trigger actions such as enabling or disabling the UI, activating spray, or switching movement modes. These controls are mainly used during development and debugging when a VR headset is not available.

The user interface is designed in world space so that it appears in front of the user within the VR environment. This allows users to read instructions and receive feedback without breaking immersion. The UI provides real-time guidance for each step, helping users understand the correct procedure.

Overall, the XR interaction design ensures that the training is interactive, realistic, and easy to follow, improving both user engagement and learning effectiveness.

Screenshots

Figure 1: Factory Environment

Figure 2: Fire Extinguisher Model

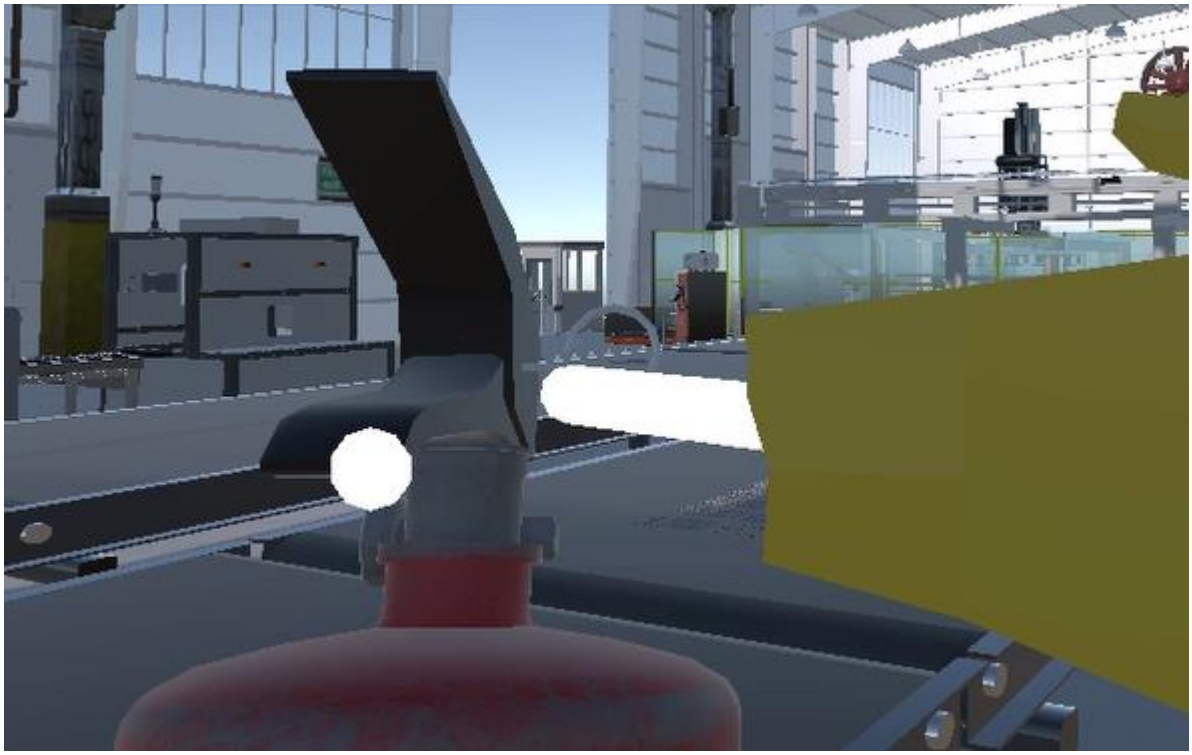
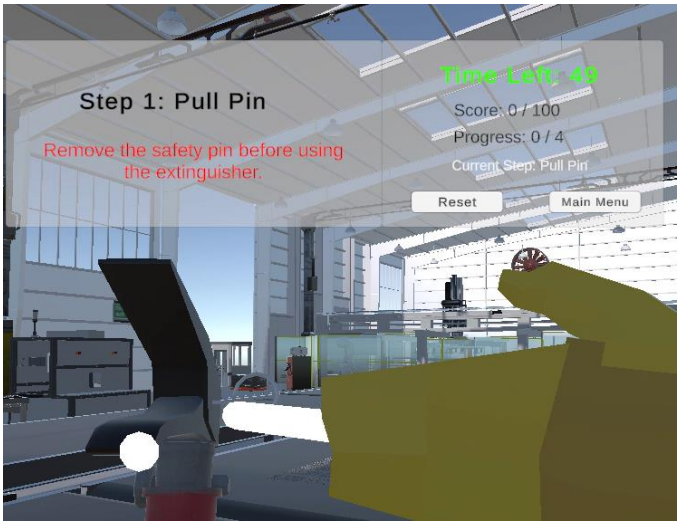


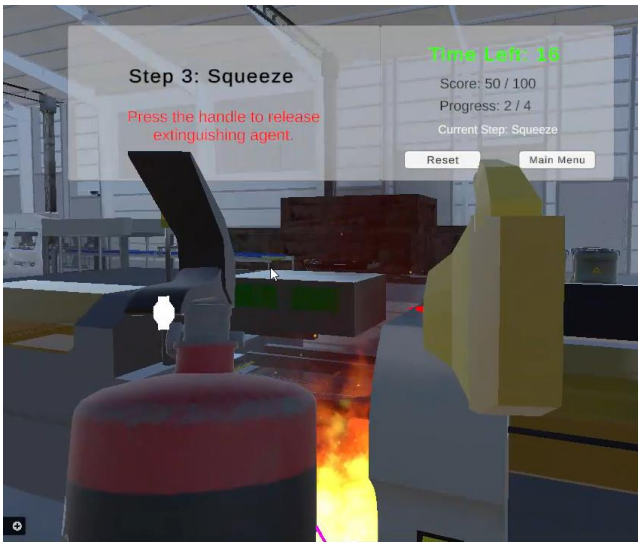
Figure 3: Pin Removal Interaction

Figure 4: Spray Particle Effect



Step1: Pull Pin

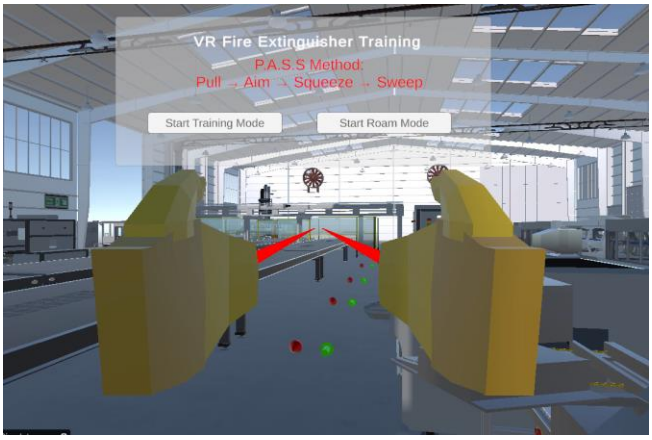
Step 2: Aim



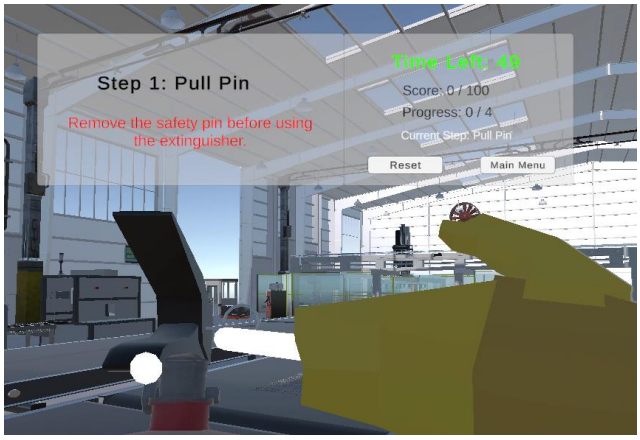
Step 3: Squeeze

Step 4: Sweep

Figure 5: Fire Extinguishing Process

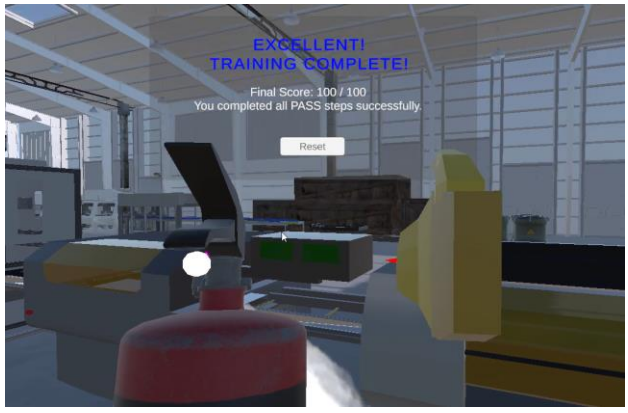


Main Menu UI

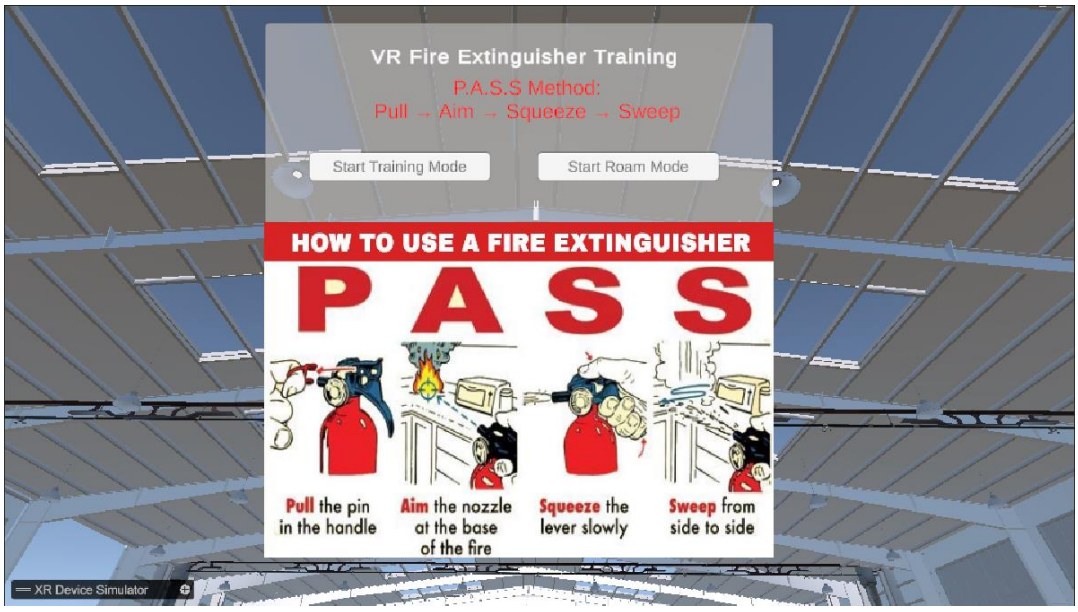


Training Mode UI

Fail Message UI



Success Message UI



Display P.A.S.S User Guideline

Figure 6: User Interface Prompts

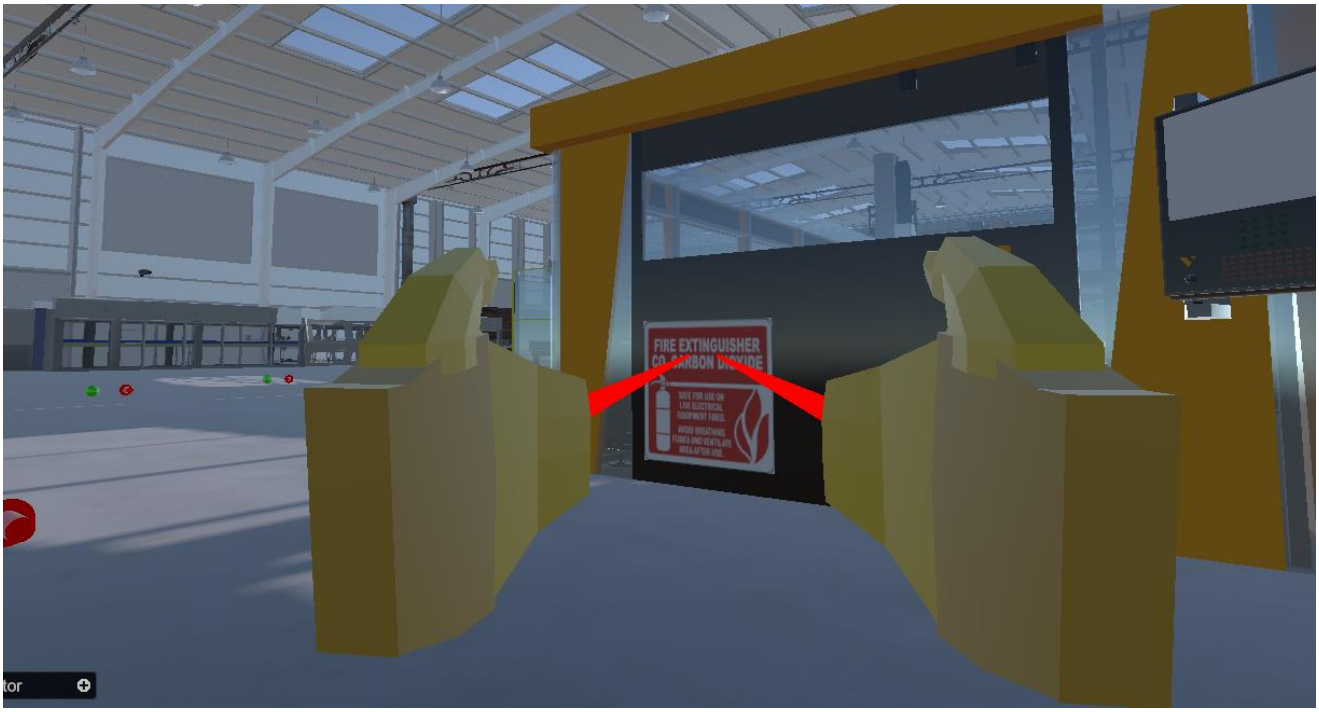


Figure 7: CO_2 Fire Extinguisher Safety Sign

Figure 8: Emergency Fire Exit Sign

Team Contribution

Task No.	Task Description	Team Member(s)
1	Fire system development (particle and intensity)	Mak Zi Qian, Ong Ti Xuan
2	Fire extinguisher setup and configuration	Mak Zi Qian, Ong Ti Xuan, Yip Siew Hui
3	XR grab interaction implementation	Mak Zi Qian, Ong Ti Xuan, Yip Siew Hui
4	Safety pin removal system (Pull step)	Ong Ti Xuan, Yip Siew Hui
5	Aiming system using raycast and line renderer	Mak Zi Qian
6	Spray activation system (Squeeze step)	Mak Zi Qian, Ong Ti Xuan, Yip Siew Hui
7	Sweep detection system (Sweep step)	Ong Ti Xuan, Yip Siew Hui
8	Fire interaction and extinguishing logic	Mak Zi Qian, Ong Ti Xuan
9	UI design (instructions, timer, score, progress)	Mak Zi Qian
10	UI panels (menu, training, success, fail)	Mak Zi Qian
11	Reset function implementation	Mak Zi Qian
12	Audio system (spray, success, fail sounds)	Mak Zi Qian
13	Enable and disable of P.A.S.S User Manual Guideline	Yip Siew Hui
14	Testing and debugging (VR and simulator)	Mak Zi Qian, Ong Ti Xuan, Yip Siew Hui
15	System integration	Mak Zi Qian, Ong Ti Xuan, Yip Siew Hui
16	Movement system (Teleportation & Snap Turn)	Yip Siew Hui
17	Safety signage integration (CO_2 extinguisher labels, emergency exit signs, industrial warning signs)	Mak Zi Qian
18	Report writing and documentation	Mak Zi Qian
19	Video Demo	Ong Ti Xuan

Asset Credits

Fire Particle Effect (Fire 001)

Author: N2 Studio VFX

Source: Unity Asset Store

Link: https://assetstore.unity.com/packages/vfx/particles/fire-explosions/fire-001-326856

License: Standard Unity Asset Store EULA

Exit Sign 1

Source: RS Components

Link: https://media.rs-online.com/image/upload/bo_1.5px_solid_white,b_auto,c_pad,dpr_2,f_auto,h_399,q_auto,w_710/c_pad,h_399,w_710/R7631793-02?pgw=1

Exit Sign 2

Source: HiClipart (original author unknown)

Link: https://www.hiclipart.com/free-transparent-background-png-clipart-iuuyx

Fire Extinguisher CO2 Sign

Source: SafetySign

Link: https://www.safetysign.com/products/20494/fire-extinguisher-co2-carbon-dioxide-sign

Description: Used as a visual reference for fire extinguisher labeling in the simulation.

P.A.S.S. Picture

Source: HASEEN HABIB

Link: https://haseenhabib.com/how-to-use-a-fire-extinguisher-pass-technique/

All assets and image are used for academic and educational purposes only.